

Precalculus Review for Calculus

ANSWER KEY



Functions and composition

- 1 Let $f(x) = \sqrt{x-1}$ and $g(x) = x^2 + 3$. Find:
- a** $(f \circ g)(x)$ $f(g(x)) = \sqrt{x^2 + 2}$.
- b** $(g \circ f)(x)$ $g(f(x)) = (x-1) + 3 = x + 2$, for $x \geq 1$.
- c** the domain of $f \circ g$ $x^2 + 2 \geq 1$ always holds, so the domain is all real numbers.
- 2 Simplify the difference quotient $\frac{f(x+h) - f(x)}{h}$ for $f(x) = x^2 - 3x$.
- $$\frac{(x+h)^2 - 3(x+h) - (x^2 - 3x)}{h} = \frac{2xh + h^2 - 3h}{h} = 2x + h - 3.$$

Lines, exponentials, and trig — the toolkit calculus builds on

- 3 Find the slope and equation of the line through $(-1, 4)$ and $(3, -2)$.
- Slope $= \frac{-2-4}{3-(-1)} = -\frac{3}{2}$; equation $y - 4 = -\frac{3}{2}(x + 1)$, i.e. $y = -\frac{3}{2}x + \frac{5}{2}$.
- 4 Solve $2e^{3x} = 14$ exactly and to three decimal places.
- $e^{3x} = 7$, so $x = \frac{\ln 7}{3} \approx 0.649$.
- 5 State the exact value of each expression, needed frequently when differentiating trig functions:
- a** $\sin \frac{\pi}{4} = \frac{\sqrt{2}}{2}$ **b** $\cos \frac{\pi}{3} = \frac{1}{2}$ **c** $\tan \frac{\pi}{6} = \frac{\sqrt{3}}{3}$